

Introduction to Renewable Energy Technologies

Renewable energy sources have become a cornerstone of the global strategy to combat climate change and reduce dependence on fossil fuels. Solar, wind, hydroelectric, and geothermal power collectively accounted for approximately 30 percent of global electricity generation in 2023, a figure that continues to rise each year.

The cost of solar photovoltaic technology has fallen by more than 90 percent over the past decade, making it the cheapest source of electricity in history in many regions. This dramatic reduction in cost has accelerated adoption across both developed and developing nations.

Solar Energy

Photovoltaic (PV) systems convert sunlight directly into electricity using semiconductor materials. The most common material is silicon, which is abundant and relatively inexpensive to produce. Modern commercial solar panels achieve efficiencies of 20 to 23 percent, while laboratory records have exceeded 29 percent.

Concentrated Solar Power (CSP) systems use mirrors or lenses to focus sunlight onto a small area, generating heat that drives a steam turbine. Unlike PV systems, CSP can store thermal energy and generate electricity after sunset, addressing the intermittency challenge.

The global installed solar capacity reached 1.6 terawatts in 2023. China alone accounts for nearly 40 percent of new installations annually. The United States, the European Union, and India are the other major markets driving growth.

Wind Energy

Wind turbines harness kinetic energy from moving air and convert it into electricity.

Onshore wind is currently the lowest-cost source of new electricity generation in most parts of the world. Turbine heights have increased substantially, with modern machines reaching hub heights of 150 meters and rotor diameters exceeding 200 meters.

Offshore wind offers higher capacity factors due to stronger and more consistent wind speeds over the ocean. The North Sea has emerged as the world's leading offshore wind region. The United Kingdom, Denmark, and the Netherlands have made substantial investments in offshore infrastructure.

Energy Storage

The intermittent nature of solar and wind power requires effective storage solutions to ensure grid stability. Lithium-ion batteries currently dominate the stationary storage

market, with costs falling sharply alongside electric vehicle adoption.

Grid-scale battery installations have surged globally, with notable deployments in Australia, the United States, and China. The Hornsdale Power Reserve in South Australia, one of the earliest large-scale installations, demonstrated that battery storage could respond to grid events faster than conventional power plants.

Long-duration storage remains a challenge. Technologies under development include flow batteries, compressed air energy storage, pumped hydro, and green hydrogen. Green hydrogen, produced by electrolysis using renewable electricity, is seen as a key solution for seasonal storage and hard-to-decarbonize industrial sectors.

Policy and Investment

Global investment in clean energy reached 1.8 trillion USD in 2023, surpassing fossil fuel investment for the first time. Government policies such as the US Inflation Reduction Act, the EU Green Deal, and China's Five-Year Plans have been instrumental in driving this capital allocation.

Carbon pricing mechanisms, including emissions trading schemes and carbon taxes, provide additional economic incentives for the transition. As of 2024, more than 70 jurisdictions have implemented or are planning carbon pricing instruments covering approximately 23 percent of global greenhouse gas emissions.